

BAXTER HEALTHCARE

R&D Solution Architecture

Detailed Design Document

R&D • Clinical Trials • Manufacturing Tech Transfer
Wymore Model M Integration | AI4RE Semantic Requirements Gateway
RDSG v2.0 | MTBH Circuit Breaker | Human-AI Teaming

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Version: 2.0 | Classification: CONFIDENTIAL — DRAFT | May 2026

1. Executive Summary

This Detailed Design Document (DDD) establishes the unified, fully integrated Holonic Semantic Architecture for Baxter Healthcare's Research & Development, Clinical Trials, and Manufacturing operations. It synthesizes and extends four foundational frameworks — the Holonic Semantic Architecture DDD, the Baxter FABRIC Solution Architecture, the Holonic Semantic Governance Architecture v2.0, and the Human-AI Teaming Framework — into a single coherent blueprint.

The architecture is organized around a seven-layer holonic model enriched by two structural innovations not present in prior documents: (1) the Wymore System Model (Model M), which provides a mathematically rigorous formal system definition governing how each layer's state space, transition functions, and performance specifications interoperate; and (2) the AI4RE Semantic Requirements Engineering Gateway (SREG), which serves as the front-end intake point for requirements originating from Marketing (Customer and Company needs), Operations (Continuous Improvement), and Compliance (regulatory lineage for PEP enforcement and dynamic governance).

Together, these elements produce an architecture that governs emergent AI capabilities through a formal semantic contract at every layer boundary, from physical lab equipment and bioreactors through clinical trial material management and into manufacturing tech transfer. Every data artifact, AI recommendation, and system state transition is governed, traced, validated, and explainable — satisfying GxP, 21 CFR Part 11, EU AI Act, and FDA AI/ML guidance simultaneously.

KEY VALUE PROPOSITIONS

Dimension	Capability	Governing Mechanism
Palantir Foundry Equivalence	Seven-layer open-standards platform at 75–80% lower TCO	FABRIC + AAS + OWL 2 + SPARQL
Wymore Model M	Formal system definition governing all layer transitions	$M = (X, Y, W, S, \Delta, \wedge, R)$ mapped to each holon
AI4RE SREG	Requirements from Marketing, Ops, and Compliance feed live governance	Structured Prompt Object → RCB → PEP
Holonic Architecture	Modular, self-governing layers composable across R&D and manufacturing	Koestler holarchy + Cagle digital orchestration
MTBH Circuit Breaker	Quantitative AI trustworthiness with automatic model suspension	Dimensional WHO/WHAT/WHEN/WHERE/WHY scoring
Human-AI Teaming	Trust calibration, fatigue detection, competency maintenance	Real-time performance monitoring + intervention
Tech Transfer	Semantic digital thread from discovery to manufacturing	SWRL rules + AAS + Teamcenter PLM + SAP ERP
GxP Compliance	Machine-executable regulatory compliance at every layer	SHACL + LKIF deontic norms + 21 CFR Part 11

2. Wymore System Model (Model M) Integration

The Wymore System Model (Model M) provides the formal mathematical foundation underlying every layer of the holonic architecture. Wymore defines a system as a seven-tuple:

$$M = (X, Y, W, S, \Delta, \Lambda, R)$$

where X is the input space, Y is the output space, W is the state space, S is the set of admissible stimulus segments, Δ is the state transition function, Λ is the output function, and R is the performance specification set. This formalism is applied at each holonic level to provide rigorous guarantees about system behavior across the R&D-to-manufacturing digital thread.

2.1 Model M Mapping Across Holonic Levels

Wymore Element	Layer 1–2 (Physical + Digital Twin)	Layer 3–4 (PEP + Knowledge Graph)	Layer 5–6 (Governance + Orchestration)	Layer 7 (Enterprise)
X — Input Space	Sensor telemetry, lab commands, clinical sample intake	AI-generated RDF event proposals, SPARQL queries	Business rules, LKIF deontic norms, SHACL shapes	Regulatory submissions, ERP orders, PLM BOMs
Y — Output Space	Physical actuations, robot motions, equipment settings	Validated RDF triples, graph updates, audit records	Governance decisions, workflow triggers, HITL alerts	Tech transfer packages, CMC dossiers, SAP BOMs
W — State Space	Equipment operational states (running, calibrating, fault)	Graph version snapshots, SHACL compliance scores	Ontology lifecycle states (draft, approved, deprecated)	Program/product lifecycle phases
S — Stimulus Segments	OPC-UA signal streams, HL7/FHIR clinical feeds	AI agent action sequences, RDSG traversal events	Requirement change requests, regulatory updates	Phase gate evidence packages
Δ — State Transition	PLC logic, SFC sequences, robotic control programs	SHACL enforcement engine, SWRL inference rules	Ontology change protocol (ORSD → KGCL), PEP rules	Architecture Decision Records (ADRs)
Λ — Output Function	Actuator commands, lab notebook entries, batch records	Compiled AI Directives, rejection records, audit trails	Governance decisions, alert notifications	Tech transfer packages, regulatory submissions
R — Performance Spec	Equipment SLAs, GxP process parameter bounds	MTBH dimensional thresholds, SHACL compliance ≥99.5%	Ontology coverage ≥85%, governance cycle time	TCO targets, regulatory compliance scores

2.2 Wymore State Transition Governance

Every state transition within the holonic architecture must satisfy the performance specifications R before it can propagate to adjacent layers. This creates a formal contract at each layer boundary that is enforced by the Policy Enforcement Point (PEP) at Layer 3. The SHACL constraint library implements the R -specification checks: any proposed state transition that violates a SHACL shape is halted at the PEP, logged to the audit trail, and routed to Human-in-the-Loop (HITL) review.

In the context of R&D operations, this means that a robot proposing a new mixing protocol (a state transition Δ in Layer 1) must generate an RDF event that passes through the Layer 3 PEP, which checks the proposed new state W' against the Layer 5 SHACL rulebook. Only if R is satisfied does the transition complete and the new operational state become published to the Knowledge Graph.

3. AI4RE Semantic Requirements Engineering Gateway (SREG)

The AI4RE Semantic Requirements Engineering Gateway (SREG) is the primary intake mechanism for all requirements that drive the architecture's governance DNA. It uses AI-assisted requirements engineering — the AI4RE tool — as a structured front-end that captures, classifies, formalizes, and traces requirements from three distinct organizational sources into the semantic governance architecture.

3.1 Requirements Source Domains

Source Domain	Representative Inputs	Formalization Output	Downstream Consumer
Marketing — Customer & Company Needs	VOC (Voice of Customer), market requirements, product specifications, competitive intelligence, patient outcome targets	OWL class definitions for product concepts; SKOS controlled vocabulary for domain terms; SWRL rules for product-performance relationships	R&D Intelligence Platform (L7); Formulation Knowledge Graph (L5); Product ontology (L5)
Operations — Continuous Improvement	Process deviation reports, OEE metrics, cycle time targets, yield improvement initiatives, equipment utilization data	SHACL shapes for process parameter bounds; SWRL temporal rules for SPC thresholds; MTBH performance targets per use case	Manufacturing Process Analytics (L7); Digital Twin (L2); PEP enforcement (L3)
Compliance — Regulatory Lineage	FDA guidance documents, ICH harmonization, GxP requirements, 21 CFR Part 11, EU AI Act, cGMP standards, SOX controls	LKIF deontic norms (obligations, prohibitions, permissions); PEP rule sets; SHACL constraint library; Audit trail requirements	PEP enforcement engine (L3); Governance framework (L5); Regulatory Intelligence Hub (L7)

3.2 AI4RE Processing Pipeline

The AI4RE tool implements a structured five-stage pipeline to transform raw organizational inputs into formal semantic requirements:

1. Ingestion & Classification — Natural language requirements from all three source domains are ingested via the AI4RE front-end interface. The tool applies NLP-based classification to identify requirement type (functional, quality, constraint, stakeholder), source domain, and applicable regulatory framework. Each requirement is assigned a unique ORSD identifier.
2. Semantic Decomposition — AI4RE decomposes compound requirements into atomic semantic units. Each unit is mapped to existing ontology classes in the FABRIC Knowledge Graph or flagged for new ontology term creation. The decomposition preserves the full traceability chain from organizational source to semantic artifact.
3. LKIF Deontic Formalization — Compliance requirements are automatically translated into LKIF deontic statements (lkif:obligation, lkif:permission, lkif:prohibition). These become first-class

governance artifacts linked to the relevant ontology classes and enforced by the PEP at Layer 3.

4. SHACL / SWRL Generation — Operational and product requirements are translated into SHACL constraint shapes and SWRL inference rules. Generated artifacts pass through the Ontology Governance Committee review workflow in VocBench 3 before deployment to the production Knowledge Graph.
5. Requirement Context Bundle (RCB) Registration — Each formalized requirement is registered as a Requirement Context Bundle component, making it available for RDSG gateway traversal. The WHO/WHAT/WHEN/WHERE/WHY dimensional tags are populated from the source domain classification. The RCB feeds directly into the PEP enforcement engine.

3.3 SREG Integration with RDSG

The Semantic Requirements Engineering Gateway is the upstream feeder of the Requirements-Driven Semantic Gateway (RDSG v2.0). Where the RDSG governs individual AI agent interactions at run-time, the SREG governs the governance DNA that the RDSG enforces — the formal source of every SHACL shape, LKIF deontic norm, and PEP rule in the system.

This creates a closed governance loop: Marketing, Operations, and Compliance requirements enter the SREG → are formalized into semantic governance artifacts → are enforced at run-time by the RDSG → generate audit evidence that feeds back to Compliance → trigger Continuous Improvement requirements back into the SREG. No governance rule exists in the system without a traceable organizational requirement source.

4. Tool Alternatives & Recommendations

The following comprehensive evaluation covers all architectural layers, incorporating alternatives for every major functional category. Final recommendations are annotated with rationale grounded in Baxter's specific requirements: GxP compliance, Siemens ecosystem integration, open-standards governance, and MTBH-governed AI trustworthiness.

4.1 Requirements Engineering & Governance Front-End

Capability	Recommended	Alternatives	Rationale
AI-Assisted Requirements Capture	AI4RE (Academic/COTS)	IBM DOORS Next, Jama Connect, Codebeamer	AI4RE provides native NLP-to-formal-specification translation; integrates with OWL ontologies; supports LKIF deontic output. DOORS and Jama require custom ontology bridges.
Ontology Lifecycle Governance	VocBench 3 + OntoDB	Protégé, TopBraid EDG, Neonto Editor	VocBench provides peer-review workflow (propose→review→approve→deploy) with Git integration and SHACL pre-deployment testing. TopBraid EDG is viable alternative with stronger enterprise support. Protégé is desktop-only.
SHACL/SWRL Authoring	Neonto Editor + VocBench	TopBraid EDG, SHACL Workbench	Neonto balances visual lifecycle management with robust OWL/SHACL support, enabling business users to manage rules without writing code. TopBraid EDG evaluated as fallback.

4.2 Execution, Orchestration & Human-in-the-Loop UI

Capability	Recommended	Alternatives	Rationale
Holon UI & Orchestration Platform	Siemens Mendix	OutSystems, Appian, Microsoft PowerApps	Native Siemens Xcelerator integration (Teamcenter/Opcenter) is critical for hardware/manufacturing tech transfer. Superior multi-experience UI for scientists. Process+ and Red Hat PAM evaluated and rejected due to weaker Siemens integration.
Workflow Engine	Mendix Workflow Engine (OOTB)	Red Hat PAM, Camunda	Standardizes task routing and GxP sign-off approvals. Native Mendix; no integration overhead.
HITL Intervention Interface	Custom React in Mendix Studio Pro	Siemens Industrial App Suite, SAP Fiori	HITL Intervention Pane requires side-by-side current/proposed state comparison with highlighted

Capability	Recommended	Alternatives	Rationale
			AI impact — best implemented as custom React widget. Mendix Studio Pro enables this without separate frontend stack.
Human Performance Monitoring	Custom Node.js microservice + PostgreSQL	Splunk, Datadog APM	Purpose-built performance monitoring service tracking cognitive load, decision accuracy, trust calibration, and fatigue scores. Custom implementation required to integrate with Mendix workflow events.

4.3 Knowledge Graph & Semantic Storage

Capability	Recommended	Alternative	Rationale
Production Triplestore	Stardog Enterprise	GraphDB Enterprise Edition	Stardog: native GraphQL interface, integrated ML inference, enterprise RBAC, sub-second SPARQL on 10B+ triples. GraphDB EE viable fallback with strong OWL 2 RL support. Neo4j rejected: lacks native RDF/OWL reasoning and SHACL validation required for GxP.
Federated SPARQL	Stardog Federated SPARQL 1.1	Apache Jena Fuseki, Oxigraph	SERVICE federation enables cross-triplestore queries spanning R&D, manufacturing, and supply chain named graphs in a single query.
Ontology Versioning	VocBench 3 + Git (ontology repo)	OntoDB, Fluent Editor	Git-backed full change history with KGCL change records. Every class, property, and axiom tracked from ORSD to deprecation.
Vector/Semantic Search	FAISS + SBERT embeddings	Weaviate, Pinecone, Qdrant	FAISS provides high-throughput ANN search indexed by domain ontology classes. Weaviate evaluated as managed alternative. FAISS preferred for on-premise air-gapped manufacturing sites.

4.4 Semantic Harmonization — Siemens AAS Layer

Capability	Recommended	Alternative	Rationale
Universal Entity Model	Siemens AAS (IEC 63278 / IDTA)	Custom OWL ontology, OSDU	AAS eliminates 6–12 months of custom ontology development via pre-built IDTA industrial ontologies, OPC-UA pharma companion specs, and ECLASS

Capability	Recommended	Alternative	Rationale
			12.0. Open international standard with no vendor lock-in.
Industrial IoT Semantic Platform	Siemens Industrial Operations X (IOX)	Azure Digital Twins, AWS IoT TwinMaker	IOX provides production-grade OPC-UA adapter for all Siemens manufacturing assets. Converts custom PLC integration into standard configuration. Critical for Opcenter MES integration.
Process Ontology	VDI/VDE 3682	ISA-88, BPMN 2.0	VDI/VDE 3682 models pharma manufacturing processes as formal models with inputs, outputs, operators. Applied to compounding, filling, lyophilization, sterilization, and packaging.
Product Classification	ECLASS 12.0 (via Siemens IOX license)	GS1, UN/SPSC	49,000+ classes and 17,000+ properties. Seed ontology for Baxter's domain model; enables cross-BL product comparison and external regulatory alignment.

4.5 Data Fabric & Integration

Capability	Recommended	Alternative	Rationale
Table Format	Apache Iceberg	Delta Lake, Apache Hudi	ACID-compliant, schema evolution, time-travel. Runs on any compute engine (avoids Databricks lock-in). Delta Lake used only as streaming landing zone.
SQL Transformation	dbt Core (self-hosted)	dbt Cloud, Informatica, Talend	Open-source, column-level lineage, built-in testing, dbt Semantic Layer as single metric source of truth. Self-hosted avoids per-seat recurring cost.
Batch ETL / Data Movement	Apache NiFi	MuleSoft, Informatica, Azure Data Factory	Visual flow-based pipeline with native SAP BAPI/RFC support. Full data provenance captured as lineage artifact. NiFi flow graphs readable by Apache Atlas.
Event Streaming	Apache Kafka + Kafka Streams	AWS Kinesis, Azure Event Hub	Distributed event streaming for Historian, SCADA, IoT, and MES events. 30-day retention enables event replay for fault investigation.
Federated Query	Trino / DuckDB	Presto, Dremio, Databricks SQL	Trino for interactive cross-BL SQL without data movement. DuckDB for edge analytics at manufacturing sites during network outages.

Capability	Recommended	Alternative	Rationale
SaaS Connectors	Airbyte (Open Source)	Fivetran, Stitch	300+ pre-built connectors. Manages CDC for relational sources. Supplements NiFi for standard vendor connectors.
API Gateway	Kong API Gateway	Apigee, AWS API Gateway	Rate limiting, mTLS, schema validation, audit logging for all external data consumers (CROs, CMOs, suppliers).
Data Quality	Great Expectations	Monte Carlo, Soda Core	200+ expectation types. Suite per domain. DQS scores block data promotion to Knowledge Graph on failure.
Metadata Catalog	OpenMetadata + Apache Atlas	Alation, Collibra, DataHub	Atlas for automated lineage from NiFi to dashboards. OpenMetadata for unified catalog with business glossary and SHACL ontological definitions.

4.6 AI & Analytics

Capability	Recommended	Alternative	Rationale
Business Intelligence	Apache Superset + dbt Semantic Layer	Tableau, Power BI, Looker	Open-source, 40+ chart types, native dbt Semantic Layer integration. All dashboards connect to governed metrics — no ad-hoc SQL. Replaces scattered Tableau/Power BI licenses (Wave 2 rationalization).
ML Lifecycle	MLflow (self-hosted)	Azure ML, SageMaker, Vertex AI	End-to-end experiment tracking, model registry, deployment. MTBH telemetry hooks at serving layer. Self-hosted on Baxter infrastructure.
Generative AI / LLM Gateway	vLLM + Ollama (self-hosted)	Azure OpenAI, AWS Bedrock	Self-hosted only — no PII or trade secrets to external API. Models: Llama 3.1 70B (general), BioMedLM (biomedical), Baxter fine-tunes (CMC/regulatory).
AI Agent Orchestration	LangGraph (multi-step agents)	AutoGen, CrewAI, Semantic Kernel	Multi-step agent framework for batch investigation, regulatory evidence retrieval, supply chain risk assessment. All agents operate over SPARQL + Iceberg + document retrieval.
AutoML / Predictive Models	FLAML	H2O AutoML, AutoSklearn	Fast, lightweight AutoML for R&D scientists without ML expertise. Batch yield prediction, stability forecasting, equipment failure prediction.

Capability	Recommended	Alternative	Rationale
Model Serving	Ray Serve	BentoML, Seldon, Triton	Distributed serving with autoscaling, model versioning, A/B splitting. MTBH telemetry hooks at inference layer.
Operational Monitoring	Grafana + ClickHouse adapter	Splunk, Datadog	Real-time manufacturing KPI dashboards. Historian and Delta Lake time-series data. Separate from Superset (Grafana = operations; Superset = analysis).
MTBH Circuit Breaker	Custom Python service (TVE proprietary)	No direct commercial equivalent	Quantitative AI trustworthiness measurement. Dimensional WHO/WHAT/WHEN/WHERE/WHY scoring. Automatic model suspension below threshold. Core TVE IP.

5. The Unified 7-Layer Holonic Architecture

The seven-layer holonic architecture organizes all design patterns, tools, governance mechanisms, and operating flows into a nested holarchic structure. Each layer is simultaneously a self-governing whole and a governed part of the layer above. The Wymore System Model governs state transitions at every layer boundary; the RDSG enforces governance protocol at every inter-layer crossing; and the MTBH Circuit Breaker monitors holonic health across all five relevancy dimensions.

The architecture operates across three operational domains simultaneously: (1) R&D Systems and Processes — experimental formulation, method development, stability studies; (2) Labs and Equipment — bioreactors, automated pipettes, lab robots, analytical instruments; and (3) Clinical Trials — investigational product management, clinical material tracking, trial site data integration.

5.1 Layer 7: Enterprise Ecosystem — The Consumers

WYMORE MODEL M: OUTPUT SPACE (Y) AND PERFORMANCE SPECIFICATIONS (R) FOR THE ENTERPRISE

Layer 7 is the enterprise consumer layer — the set of systems and processes that consume the 'finished data packages' produced by the six lower layers. In Wymore terms, this layer defines the Y (output space) and R (performance specifications) for the entire holarchy. Every design decision in Layers 1–6 is ultimately traceable to a business outcome defined here.

5.1.1 Enterprise System Inventory

System	Category	Role in Architecture	Integration Mode
SAP S/4HANA	ERP	Authority for BOM, cost center, supply chain master data. Consumes tech transfer packages from R&D as manufacturing BOMs.	Read-only Ph1–2; write-back Ph3 via FABRIC canonical identifiers (FCIDs)
Siemens Teamcenter PLM	PLM	Product lifecycle, DHF, ECO/ECR, multi-level BOM. Receives unified Bill of Process (BOP) from tech transfer flow.	AAS submodel read from Teamcenter BOM; Mendix pushes harmonized BOP on transfer
Veeva Vault QMS/RIM	Quality/Regulatory	CAPA, deviations, change control, SOPs, submissions, labeling. Authority for all regulated document content.	Read-only; Regulatory Intelligence Hub queries via HL7/FHIR bridge and REST API
Regulatory Intelligence Hub	L7 Capability App	CMC traceability from development batch to regulatory submission. eCTD section population assistance via LangGraph Regulatory Evidence Agent.	Composed from L5 Knowledge Graph + L6 LangGraph agents + Veeva Vault
Quality & Compliance Command Center	L7 Capability App	Integrated CAPA, deviation trending, audit readiness scoring, predictive quality risk model.	Composed from L5 SHACL compliance data + L4 dbt metrics + L6 Grafana/Superset

System	Category	Role in Architecture	Integration Mode
R&D Intelligence Platform	L7 Capability App	Cross-BL formulation query, ELN harmonization view, failed experiment search, patent landscape.	Composed from L5 Knowledge Graph + L6 vLLM RAG + L4 Iceberg datasets

5.1.2 AI4RE Feed to Enterprise Layer

The SREG provides continuous requirements intelligence to Layer 7 capability applications. Marketing requirements (product performance targets, patient outcome specifications) are formalized as OWL classes that drive the R&D Intelligence Platform's knowledge graph queries. Operations CI requirements are translated into dbt Semantic Layer metric definitions consumed by Manufacturing Process Analytics. Compliance requirements become SHACL constraints enforced across all L7 capability applications' audit trails.

5.2 Layer 6: Holon Orchestration & UI — The Interface

WYMORE MODEL M: OUTPUT FUNCTION (Λ) — USER-FACING OUTPUTS AND HUMAN GOVERNANCE DECISIONS

Layer 6 is the primary interaction and orchestration layer. It provides scientists, lab managers, clinical operators, and manufacturing engineers with custom, role-based dashboards and workflow management. Critically, it is the layer where Human-in-the-Loop (HITL) governance decisions are made, human performance is monitored, and trust calibration is enforced.

5.2.1 Component Inventory

Out-of-the-Box (OOTB) Mendix Components

- Data Hub / OData Connectors — Virtualizes ERP/PLM data without replicating into R&D space. Provides live SAP BOM and Teamcenter BOP views in R&D dashboards.
- Workflow Engine — Standardizes task routing and multi-step approval processes for GxP sign-offs. Every approval generates a 21 CFR Part 11 compliant electronic signature record.
- AI Agent Task Widget — Monitors the active status and decision trees of deployed LangGraph AI agents. Displays MTBH scores per agent and circuit breaker status.
- Role-Based Access Control (RBAC) — Enforces ontological role definitions from the Knowledge Graph. Access permissions derived from SPARQL queries against the agent registry, not manual permission matrices.

Custom React Components (Built in Mendix Studio Pro)

- SHACL Validation Viewer — Translates raw SHACL validation reports into a traffic-light UI (Green/Yellow/Red) for R&D managers and QC scientists to spot compliance violations. Includes drill-down to the specific constraint and affected triple.
- Ontology Context Explorer — A visual tree-node widget allowing scientists to explore how a specific clinical material, formulation component, or lab process relates to downstream supply chain, regulatory, and manufacturing elements. Traverses the L5 Knowledge Graph via SPARQL queries.
- HITL Intervention Pane — Side-by-side comparison tool activated when an AI agent proposes a state change. Shows 'Current State' vs. 'Proposed AI State' with highlighted Wymore model impacts (ΔW, ΔΛ). Requires logged justification before override (satisfies 21 CFR Part 11).

- **AI4RE Requirements Dashboard** — Front-end for the SREG. Displays the live requirements traceability matrix from Marketing/Ops/Compliance sources to formalized SHACL/LKIF artifacts. Enables requirement owners to review, approve, and track ORSD change requests.
- **Clinical Trial Material Tracker** — Specialized dashboard for investigational product chain-of-custody, sample genealogy, and protocol deviation management. Integrated with HL7/FHIR bridge for trial site data.
- **Human Performance Monitor** — Real-time display of knowledge engineer cognitive state (fatigue score, trust calibration status, decision accuracy trend). Triggers mandatory break enforcement and workload rebalancing via integration with the Human-AI Teaming service.
- **Wymore State Dashboard** — Visualizes the current W (state space) across all active holons in each domain. Displays pending state transitions Δ and their Wymore performance specification compliance status.

5.2.2 Human-AI Teaming Integration at Layer 6

The Human-AI Teaming Framework is implemented as a Layer 6 microservice layer that wraps all HITL interactions. It monitors seven dimensions of human performance and provides corresponding interventions:

Human Performance Dimension	Detection Mechanism	Layer 6 Intervention	Governance Record
Automation Complacency	Decision velocity > 2.5x baseline; accuracy drop > 15%	Warning banner; forced review checklist before accept button enables	KGCL record flags decisions made during complacency alert state
Alert Fatigue	Override rate > 20%; queue clearance velocity spikes	Alert prioritization by MTBH severity; adaptive rate limiting on low-priority flags	Audit log captures all override events with engineer cognitive state at time
Trust Mis-calibration	Acceptance rate vs. actual accuracy gap > 15% per confidence bucket	Personalized calibration warning before each decision in mis-calibrated range	Trust calibration report fed to domain governance team weekly
Skill Degradation	Manual mapping exercise score > 15% below baseline	Automatic training scheduling; follow-up assessment in 7 days	Competency assessment record in Performance Database
Cognitive Overload	Fatigue score > 0.7 (session duration, accuracy, decision speed composite)	Mandatory 15-minute break enforcement; task complexity reduction	Session fatigue log; workload management audit
Mode Confusion	Agent activation state not visible to engineer during query	Mode awareness panel shows active agents, applied context, and missing coverage	Query context snapshot stored with validation decision record
Production Pressure	Team pressure score > 0.7; > 50% team showing rush behavior	Management dashboard alert; halt non-critical work; 30–50% target reduction	Safety culture incident record; management intervention log

5.3 Layer 5: Semantic Governance — The Rulebook

WYMORE MODEL M: PERFORMANCE SPECIFICATIONS (R) AND STATE TRANSITION CONSTRAINTS

Layer 5 is the central repository for all business logic, regulatory constraints, and data definitions. It defines the boundaries of what is 'legal' within any given holon. In Wymore terms, this layer operationalizes the R (performance specifications): every constraint a state transition Δ must satisfy before it is permitted. The SREG feeds new R-specifications into Layer 5 continuously as Marketing, Ops, and Compliance requirements are formalized.

5.3.1 Semantic Governance Component Inventory

Component	Tool	Holonic Function	GxP / Regulatory Linkage
OWL 2 Domain Ontology	VocBench 3 + Stardog	Knowledge holon identity and semantic class definitions. Extends BFO with IDTA AAS-derived classes and Baxter pharma domain classes.	ICH Q10 pharmaceutical quality system; FDA product classification alignment
SHACL Constraint Library	Neonto Editor + VocBench	200+ shapes enforcing GxP data integrity across the graph. Batch-specification relationships, supplier qualification temporal rules, batch genealogy completeness, regulatory linkage requirements.	21 CFR Part 211 (GMP), ICH Q7 (API), FDA 21 CFR Part 11 (electronic records)
SWRL Temporal Rules	Stardog SWRL engine	Automated inference: specification expiration, supplier qualification windows, stability interval enforcement, retest date calculation, tech transfer parameter mapping.	ICH Q1A (stability), ICH Q7 (API qualification), ICH Q10 (change management)
LKIF Deontic Norm Library	Custom OWL/LKIF extensions	Formal prescriptive governance: obligations (batch pH must be between 7.0–7.4), permissions ($\pm 2\%$ variance allowed), prohibitions (misrepresenting clinical data), violations (trip Circuit Breaker), sanctions (HITL mandatory).	SOX compliance, 21 CFR Part 11, EU AI Act Article 9 risk management
Ontology Requirements Spec Docs (ORSD)	VocBench + Git	Design-time specification of every knowledge holon's governance contract. Holonic equivalent of MBSE system requirements baseline.	Traced to SREG source requirements (Marketing, Ops, Compliance)
Knowledge Graph Change Language (KGCL)	Stardog + custom tracker	Run-time change records: WHAT (change), WHY (rationale), WHO, WHEN. Every change to semantic asset captured with full provenance.	21 CFR Part 11 audit trail; FDA electronic records compliance

Component	Tool	Holonic Function	GxP / Regulatory Linkage
Vocabulary Management	VocBench 3 SKOS	Controlled vocabulary per domain (product, process, organizational, regulatory) with lifecycle: proposed→review→approved→stable→deprecated.	ICH M2/M4 terminology harmonization; FDA submission vocabulary
External Ontology Alignments	OWL owl:equivalentClass mappings	IDTA/ECLASS (manufacturing), SNOMED CT (clinical), NCI Thesaurus (regulatory), MeSH (biomedical), FIBO (financial compliance), SysML v2/KerML (MBSE).	Cross-standard regulatory traceability; CDISC for clinical data

5.3.2 Governance Workflow — Requirement to Constraint

The full lifecycle from organizational requirement to enforced semantic constraint follows the SREG-to-SHACL pipeline:

1. Marketing requirement entered in AI4RE SREG: 'Investigational product pH must remain between 6.8–7.2 during clinical trial administration.'
2. AI4RE classifies as Quality Constraint; maps to existing OWL class ClinicalBatch; generates draft SHACL shape ph_range_constraint with minInclusive 6.8, maxInclusive 7.2.
3. Draft SHACL shape submitted to Ontology Governance Committee via VocBench workflow. Quality Manager and R&D scientist review. LKIF obligation lkif:obligation(ClinicalBatch, pHInRange_6.8_7.2) attached.
4. Approved shape deployed to Stardog production Knowledge Graph. KGCL change record created. ORSD updated with effective date and regulatory reference (ICH Q6A).
5. SHACL shape is now active in Layer 3 PEP enforcement engine. Any AI agent or lab system proposing a batch state transition that would set pH outside range is automatically blocked and routed to HITL.

5.4 Layer 4: Knowledge Graph — The Digital Thread

WYMORE MODEL M: STATE SPACE (W) — ALL ENTITY STATES ACROSS THE R&D-TO-MANUFACTURING THREAD

Layer 4 is the semantic intelligence core and the living, reasoned, temporal model of Baxter's operational reality. It stores all entities as typed nodes with semantically defined, validated, and reasoned relationships — enabling cross-domain queries that no individual source system can answer. In Wymore terms, it is the W (state space): the complete set of valid system states across all holons at all points in time.

5.4.1 Knowledge Graph Architecture

Named Graph	Domain Scope	Primary OWL Classes	SHACL Coverage
R&D Graph	Formulation, methods, ELN,	FormulationDevelopmentBatch, StabilityStudy,	50+ shapes: formulation-stability linkage,

Named Graph	Domain Scope	Primary OWL Classes	SHACL Coverage
	stability, analytical results	AnalyticalMethod, ExperimentalFormulation, ActivePharmaceuticalIngredient, ExcipientMaterial	method validation completeness, ELN entry integrity
Clinical Trials Graph	Investigational products, trial materials, CRO feeds, CDISC	ClinicalTrialMaterial, InvestigationalMedicinalProduct, ClinicalBatch, TrialSite, CRODataFeed, BioanalyticalResult	40+ shapes: material chain-of-custody, protocol deviation classification, CDISC-SEND/SDTM alignment
Manufacturing Graph	Batches, equipment, processes, genealogy, CPV	ManufacturingBatch, ManufacturingSite, Equipment, ManufacturingProcess, BatchRecord, SupplierOrganization	60+ shapes: batch genealogy completeness, equipment qualification temporal rules, CPV specification linkage
Supply Chain Graph	Suppliers, materials, logistics, CMO/CRO	Supplier, RawMaterial, MaterialLot, CMO, LogisticsSegment, QualificationRecord	30+ shapes: supplier qualification windows, concentration risk, lead time SLAs
Regulatory Graph	Dossiers, specifications, change control, submission linkage	RegulatoryDossier, CMCSpecification, ChangeControlRecord, eCTDSection, RegulatorySubmission	25+ shapes: specification version constraints, submission completeness, dossier-batch traceability
Governance Graph	SHACL shapes, LKIF deontic norms, ORSD, KGCL records	SHACLShape, LKIFObligation, ORSD, KGCLChangeRecord, GovernanceDecision	Integrity checking of governance artifacts themselves

5.4.2 Digital Thread Implementation

The Knowledge Graph implements the digital thread from Discovery to Production as a continuous chain of semantically typed relationships. Each stage of the product lifecycle is a connected subgraph, and SWRL rules enable automated traversal across boundaries:

- Discovery → Lead Selection: ExperimentalFormulation → hasStabilityProfile → StabilityStudy → producesResult → AnalyticalResult → meetsSpecification → DevelopmentSpecification
- Development → Clinical: DevelopmentBatch → scaledTo → ClinicalBatch → usedIn → ClinicalTrialMaterial → testedAt → TrialSite → producesData → ClinicalOutcome
- Clinical → Manufacturing: ClinicalBatch → techTransferTo → ManufacturingBatch → producedAt → ManufacturingSite → usesEquipment → Equipment → hasQualification → QualificationRecord
- Manufacturing → Regulatory: ManufacturingBatch → documentedIn → BatchRecord → supportsDossierSection → eCTDSection → partOf → RegulatoryDossier → submittedTo → RegulatoryAgency

5.5 Layer 3: Policy Enforcement Point (PEP) & AI Edge — The Gateway

WYMORE MODEL M: STATE TRANSITION FUNCTION (Δ) — ALL TRANSITIONS GOVERNED AT THIS LAYER

Layer 3 is the governance gateway. All system-to-system or AI-generated process changes must pass through this layer to be validated against the Layer 5 rulebook before reaching physical equipment (Layer 1) or being committed to the Knowledge Graph (Layer 4). In Wymore terms, this layer operationalizes Δ (the state transition function): no state transition completes without passing the Layer 5 R-specification checks enforced here.

This is also the RDSG v2.0 gateway holon — the six-stage holarchic traversal protocol that governs every AI agent interaction, requirement-driven semantic operation, and cross-domain data exchange.

5.5.1 RDSG v2.0 Six-Stage Traversal

Stage	Function	Wymore Mapping	Key Tools
Stage 0: RCB Assembly	Requirement Context Bundle assembled from AI4RE SREG. Incoming request decomposed into WHO/WHAT/WHEN/WHERE/WHY dimensions. Appropriate domain holons activated.	X (input) $\rightarrow$ initial W (state) instantiation	AI4RE SREG, Structured Prompt Object parser, Domain Holon Registry
Step 1: SHACL Enforcement	All activated domain holons' SHACL constraints checked against the proposed state transition payload. Violations flagged with severity classification.	R check: does proposed Δ satisfy SHACL invariants?	Stardog SHACL engine, SHACL constraint library (200+ shapes)
Step 2: SWRL Temporal Rules	Temporal SWRL rules applied: stability intervals, supplier qualification windows, specification expiration dates, retest date calculations.	Temporal W validation: is W' valid at time T?	Stardog SWRL engine, KGCL timestamp governance
Step 3: PEP Compliance Evaluation	Six PEP domains evaluated: Compliance, Risk, Legal, Security, Data Governance, AI Governance. Each corresponds to a LKIF deontic norm set from the Layer 5 Governance Rulebook.	R check: does Δ violate any LKIF obligation/prohibition?	LKIF deontic norm library, OPA Policy Engine, PEP rule sets
Step 4: STPA/STAMP UCA Coverage	Unsafe Control Action (UCA) coverage verified. Irreversible actions detected and suspended at Simplex Action Gateway pending human authorization.	Safety holonic invariant: UCA boundary enforcement	STPA/STAMP UCA specification, Simplex Action Gateway, HITL trigger
Step 5: Compiled AI Directive	Governance outputs synthesized into behavioral directive. Agent receives governed context — not just instructions but	Λ (output function): produces Y from validated W'	LangGraph agent runtime, Ray Serve, governed prompt assembly

Stage	Function	Wymore Mapping	Key Tools
	semantic authority for permitted action.		
Service Facade: Delivery Gate	Agent response validated against AcceptanceCriteria before leaving governance holarchy. MTBH score updated. Audit record committed to Governance Graph.	Y validated against R before release	MTBH Circuit Breaker, Stardog audit graph, Apache Atlas lineage

5.5.2 PEP Domain Specifications

The six PEP governance domains each enforce a distinct class of governance rules, derived from SREG-formalized requirements:

PEP Domain	Rule Source	Enforcement Mechanism	HITL Trigger
Compliance PEP	21 CFR Part 11, GxP, ICH Q10, EU AI Act	LKIF obligations mapped to SHACL shapes. All batch state transitions validated against GMP process parameter bounds.	Any SHACL violation on a GxP-critical parameter → immediate HITL with compliance officer notification
Risk PEP	Risk register, FMEA/FMECA outcomes, STPA UCAs	Risk scoring per proposed action. UCA boundary enforcement via STPA/STAMP analysis integrated into Wymore Δ specification.	Risk score > threshold or UCA match → Simplex Action Gateway suspension + HITL
Legal PEP	LKIF deontic norms, SOX controls, IP agreements, CMO contracts	Prohibition enforcement: actions that would violate IP agreements, data sharing agreements, or SOX controls blocked at this layer.	Legal PEP violation → automatic block + Legal team notification via Mendix workflow
Security PEP	OPA policies, RBAC ontological role definitions, zero-trust architecture	Attribute-based access control enforced by querying the agent registry. Zero-trust: every agent interaction requires authenticated context.	Unauthorized access attempt → immediate block + IT Security audit event
Data Governance PEP	Data contracts (ISO 8000), data quality SLAs, DQS thresholds	DQS below threshold blocks graph promotion. Data contract SLAs enforced before data crosses domain boundaries.	DQS < 80% for critical dataset → data held in staging; data steward notified
AI Governance PEP	MTBH thresholds (per use case risk class), EU AI Act Article 9, FDA AI/ML guidance	MTBH Circuit Breaker: model suspended when confidence falls below use-case-specific threshold. Composition Theorem:	MTBH trip → automatic failover to last stable model + HITL notification with model health report

PEP Domain	Rule Source	Enforcement Mechanism	HITL Trigger
		multiplicative fidelity tracked across multi-model pipelines.	

5.5.3 MTBH Circuit Breaker — Dimensional Health Monitoring

The MTBH (Mean Time Between Hallucinations) Circuit Breaker monitors holonic health across the five WHO/WHAT/WHEN/WHERE/WHY relevancy dimensions. Each dimension monitors a specific holonic level, enabling targeted remediation without systemic shutdown:

MTBH Dimension	Holonic Level Monitored	Measurement	Trip Action
MTBH_WHO	Persona-holon: identity context of the initiating agent/user	Fidelity of persona-level routing: is the correct domain agent invoked for this user context?	Re-route to correct domain agent; flag persona registry inconsistency
MTBH_WHAT	Concept-holon: specific knowledge artifact being accessed	OWL class resolution fidelity: does the AI's concept resolution match the authoritative ontology?	Suspend concept resolution; route to Knowledge Graph truth; update vector DB embeddings
MTBH_WHEN	Temporal-holon: time context of the interaction	Temporal anchoring fidelity: are SWRL temporal rules correctly constraining responses to current specification versions?	Re-apply SWRL temporal rules; flag stale knowledge holon; trigger KGCL staleness record
MTBH_WHERE	Domain-holon: component ontology boundary	Domain-boundary routing fidelity: is data from the correct named graph domain?	Re-route to correct domain graph; flag cross-domain contamination; domain governance team alert
MTBH_WHY	Intent-holon: governance purpose of the interaction	Intent-governance fidelity: does the AI's output align with the LKIF deontic purpose of the query?	Block output; route to HITL for intent verification; KGCL violation record

5.6 Layer 2: Digital Twin & Connectivity — The Translation Layer

WYMORE MODEL M: STIMULUS SEGMENTS (S) — ADMISSIBLE INPUT SEQUENCES FROM PHYSICAL WORLD

Layer 2 converts raw machine telemetry (temperature, pressure, pH, viscosity, flow rate) and legacy API responses into RDF triples that feed the Knowledge Graph at Layer 4. In Wymore terms, this layer defines S (the set of admissible stimulus segments): what input sequences from Layer 1 physical holons are valid inputs to the governance architecture. Only stimulus segments that can be typed as RDF triples conformant with the Layer 5 ontology are admitted.

5.6.1 Digital Twin & Connectivity Component Inventory

Component	Tool	Function	Baxter Application
Siemens MindSphere / IOX	Siemens Industrial Operations X	Cloud-based IIoT platform providing semantic context for operational data. Maps PLC/SCADA signals to AAS submodel attributes — raw sensor values become semantically typed measurements.	All Baxter manufacturing assets; lab equipment at R&D pilot facility; OPC-UA signal normalization
OPC-UA Adapters	Siemens IOX OPC-UA Adapter	Production-grade OPC-UA adapter for SCADA, DCS, PLCs, and SIMATIC automation. Eliminates custom PLC integration — standard configuration exercise.	Bioreactors, autoclaves, lyophilizers, filling lines, environmental monitoring systems
AAS Shell Management	Siemens AAS SDK (Java/Python)	Every Baxter asset — equipment, batch, material, supplier, facility — is represented as an AAS with unique global identifier and typed submodels. Site-agnostic semantic normalization.	Bioreactor at Site A is ontologically equivalent to Bioreactor at Site B regardless of vendor or configuration
Lab API Connectors	Apache NiFi + Airbyte	REST and MQTT API integration for lab instruments (spectrophotometers, particle size analyzers, rheometers, Karl Fischer titrators). Converts instrument output to RDF events.	LabVantage LIMS, Benchling ELN, IDBS ELN, Waters EMPOWER chromatography system
HL7/FHIR Bridge	Custom NiFi processor	For clinical trial product lines: HL7 v2 and FHIR R4 message transformation to FABRIC canonical format. Supports pharmacovigilance data flows and clinical outcome data feeds.	CRO data feeds, trial site electronic data capture (EDC), bioanalytical lab results
Kafka Streaming	Apache Kafka + Delta Lake	Real-time event streaming for Historian, SCADA, IoT, and MES events. Delta Lake landing zone with ACID guarantees before promotion to Iceberg governed tables.	PI Historian time-series, real-time bioreactor telemetry, environmental monitoring alerts
R2RML Mapping Pipeline	Ontop + RDFLib	Automated R2RML generation: schema extraction → RDF triple generation → SHACL validation → Knowledge Graph ingest. Brings legacy relational data into holarchy as bottom-up holons.	Legacy LIMS data (LabVantage, IDBS), MES batch record relational tables, SAP operational data

5.6.2 Lab Equipment Digital Twin Specifications

Equipment Class	AAS Submodel Templates	Semantic Normalization	OPC-UA Tags → RDF Properties
Bioreactors (cell culture, fermentation)	NameplateV2, TechnicalData, BatchRecord, ProcessPhaseLog	BioReactor → rdf:type → ManufacturingEquipment; hasCapacity, operatesAt, currentPhaseState	Temperature → hasTemperature (°C, xsd:double); pH → hasPhValue; DO → hasDissolvedOxygen; Agitation → hasImpellerSpeed
Automated Pipettes / Liquid Handlers	NameplateV2, TechnicalData, LiquidTransferRecord	LiquidHandlingSystem → rdf:type → LabRobot; hasVolumeCalibratedRange; operatesIn → LabEnvironment	VolumeDispensed → hasDispensedVolume; Accuracy → hasCVPercent; TipType → usesTipType
Lab Robots (synthesis, analysis)	NameplateV2, TechnicalData, ProtocolExecutionLog	LaboratoryRobot → rdf:type → LabEquipment; executesProtocol → LabProtocol; hasMaintenanceRecord	ArmPosition → hasRobotArmPosition; GripForce → hasGripForce; ProtocolStep → executesStep
Analytical Instruments (HPLC, MS, NMR)	NameplateV2, TechnicalData, AnalyticalMethodRecord, CalibrationRecord	AnalyticalInstrument → rdf:type → LabEquipment; usesMethod → AnalyticalMethod; hasCalibration	ChromatogramPeak → hasPeakArea; RetentionTime → hasRetentionTime; Purity → hasPurityPercent
Environmental Sensors	NameplateV2, EnvironmentalMonitoringRecord	EnvironmentalSensor → rdf:type → MonitoringSensor; monitors → LabEnvironment; hasAlertThreshold	Temperature → hasAmbientTemperature; RH → hasRelativeHumidity; ParticleCount → hasParticleCount
Clinical Material Storage Units	NameplateV2, ColdChainRecord, ChainOfCustodyRecord	StorageUnit → rdf:type → ClinicalMaterialStorage; stores → ClinicalTrialMaterial; hasTemperatureLog	StorageTemp → hasStorageTemperature; AlarmEvent → hasTemperatureExcursionEvent; Integrity → hasPackageIntegrity

5.7 Layer 1: Physical Holons — The Execution Layer

WYMORE MODEL M: INPUT SPACE (X) — PHYSICAL INPUTS TO THE GOVERNED SYSTEM

Layer 1 constitutes the physical execution environment: the bioreactors, automated pipettes, lab robots, raw clinical materials, environmental sensors, manufacturing equipment, and clinical trial site assets. In Wymore terms, these define X (the input space): the physical reality that the architecture receives, governs, and translates into the semantic fabric of the holarchy.

5.7.1 Physical Holon Classification

Physical Holon Category	Assets	Governance Interface	Key Constraints (SHACL/LKIF)
R&D Lab Equipment	Bioreactors (5L–200L), automated synthesis robots, HPLC/MS/NMR instruments, particle size analyzers, Karl Fischer titrators, dissolution baths, centrifuges	OPC-UA → Layer 2 MindSphere/IOX → AAS SubmodelTechnicalData	LKIF obligation: calibration certificate must be current; SHACL: instrument must have valid qualification status before accepting analytical results
Automated Pipettes & Liquid Handlers	Hamilton Nimbus, Tecan EVO, Biomek i7; capillary electrophoresis systems; dissolution sampling robots	Instrument API → NiFi → AAS LiquidTransferRecord	SHACL: dispensed volume must be within $\pm 0.5\%$ of target; LKIF: liquid handler tip lot must have traceability record
Lab Robots	Collaborative robots (UR10, KUKA LBR), synthesis automation platforms, colony pickers, plate handlers	OPC-UA / MQTT → IOX → RDF ProtocolExecutionLog	STPA UCA: robot must halt if human enters safety zone; SHACL: robot protocol must have validated version before execution
Clinical Trial Materials	Investigational medicinal products (IMPs), reference standards, biological samples, comparator drugs, placebos, cold-chain materials	Chain-of-custody API → HL7/FHIR bridge → ClinicalTrialMaterial AAS	LKIF obligation: IMP chain of custody must be unbroken; SHACL: storage temperature log must have no excursions; LKIF prohibition: expired IMP must not be dispensed
Environmental Control Systems	HVAC controllers, particulate monitors, pressure differential sensors, temperature/humidity sensors, UV-C sterilization units	OPC-UA / MODBUS → IOX → EnvironmentalMonitoringRecord	SHACL: cleanroom classification must be maintained; LKIF obligation: alert must be generated within 15 minutes of limit exceedance
Raw Clinical Materials	APIs, excipients, packaging materials, solvents, growth media, biological starting materials	Goods receipt scan → SAP BAPI/RFC → AAS SubmodelMaterialClassification	SHACL: material must have CoA linked before use; LKIF obligation: supplier must be currently qualified; LKIF prohibition: rejected material must not enter production

6. Operating Flows — R&D, Clinical Trials & Tech Transfer

The following operating flows describe how modular controls are managed, scaled, and orchestrated across Baxter's R&D operations. Each flow is governed by the Wymore Model M state transition specification, traverses the RDSG gateway, and generates audit evidence traceable to SREG-formalized requirements. All flows enable emergent capabilities — new best practices propagated through the digital thread — while preserving governance invariants.

6.1 Flow A: Dynamic Lab Execution — Emergent R&D Capabilities

EMERGENT CAPABILITY: ROBOT DISCOVERS OPTIMAL MIXING PROTOCOL

This flow demonstrates how emergent AI capabilities at the lab equipment level are safely governed, validated, and propagated across the R&D knowledge ecosystem without compromising GxP integrity.

Step	Actor	Action	Wymore Mapping	Governance Mechanism
1	Lab Robot (Layer 1)	Robot executing new mixing protocol uses local ML to adjust impeller speed based on real-time viscosity data (5L bioreactor, process parameter optimization)	$X \rightarrow \Delta$: physical input triggers local state transition	OPC-UA stream $\rightarrow$ IOX $\rightarrow$ AAS ProcessPhaseLog
2	Digital Twin (Layer 2)	IOX maps robot's adjusted parameter to AAS submodel: hasImpellerSpeed updated. Parameter change packaged as RDF event with equipment ID, timestamp, batch ID, and proposed new value	X translated into S : admissible stimulus segment created	AAS BatchRecord submodel update; KGCL timestamp recorded
3	PEP (Layer 3)	Edge PEP intercepts RDF event. RDSG Stage 0: RCB assembled. WHO = lab robot agent, WHAT = ImpellerSpeed, WHEN = current batch phase, WHERE = R&D graph, WHY = yield optimization intent	Δ pre-check: is this transition admissible?	RDSG RCB Assembly; all five MTBH dimensions populated
4	PEP (Layer 3)	RDSG Steps 1–4: SHACL check (impeller speed within validated range), SWRL temporal check (batch phase permits this)	R check: does proposed W' satisfy all performance specs?	SHACL validation; SWRL inference; LKIF obligation check; UCA boundary

Step	Actor	Action	Wymore Mapping	Governance Mechanism
		adjustment), PEP compliance (GxP boundary), STPA UCA check (no safety violation)		
5a — Pass	PEP (Layer 3)	Validation passes. Compiled AI Directive issued. Robot's adjusted parameter confirmed as permitted emergent behavior	Δ approved: $W \rightarrow W'$ transition permitted	RDSG Service Facade AcceptanceCriteria met; audit record committed
5b — Fail	PEP (Layer 3)	SHACL violation detected (e.g., proposed 450 RPM exceeds validated 400 RPM maximum). Circuit Breaker holds the change. Mendix HITL workflow triggered.	Δ blocked: W' violates R; HITL required	SHACL violation report; Mendix HITL Intervention Pane activated; KGCL violation record
6 (Pass path)	Knowledge Graph (Layer 4)	Validated new process parameter written to R&D Named Graph as new best practice. SWRL rule adds it to permitted parameter space for this protocol version.	W' committed: new state space available to other holons	Stardog graph update; RDF triple with provenance; KGCL change record
7 (Pass path)	Governance (Layer 5)	New permitted parameter published to domain ontology via KGCL. Other lab robots and AI agents immediately access the updated permitted behavior.	R updated: new performance specification added	VocBench ontology update; ORSD revision; SHACL constraint update if range expanded
8 (Pass path)	Orchestration (Layer 6)	R&D scientists see new best practice surfaced in Mendix dashboard. Ontology Context Explorer shows updated parameter space. MTBH scores for the robot agent updated.	Y delivered: governed output to human consumers	Mendix notification; MTBH dashboard update; Superset metric update

6.2 Flow B: Clinical Trial Material Management

CLINICAL MATERIAL CHAIN-OF-CUSTODY, PROTOCOL COMPLIANCE & CDISC TRACEABILITY

This flow governs the complete lifecycle of investigational medicinal products (IMPs) through clinical trials — from investigational product release through site dispensing, patient administration, sample

collection, and bioanalytical data return. It enforces chain-of-custody, protocol compliance, and CDISC traceability at every step.

Stage	Action	Clinical Governance	Wymore / RDSG Control
IMP Release Authorization	Quality Manager authorizes IMP release in Mendix HITL workflow. SHACL validates: CoA present, stability data current, supplier qualification active, storage conditions met, regulatory status confirmed.	21 CFR Part 312 IMP release; ICH Q10 batch release	Layer 3 PEP compliance check; LKIF obligation: all release criteria must be satisfied; Simplex Action Gate holds release pending HITL sign-off
Cold Chain Shipment Tracking	AAS ColdChainRecord updated with GPS + temperature logger data in real-time via Kafka streaming. Any temperature excursion auto-triggers HITL alert.	ICH Q1A cold chain integrity; SHACL: storageTemp within $-20^{\circ}\text{C} \pm 5^{\circ}\text{C}$	Layer 2 IoT stream → Kafka → Delta Lake; Layer 3 SHACL real-time validation; MTBH_WHEN monitors temporal freshness
Site Receipt & Chain of Custody	Trial site scans IMP package. HL7/FHIR message generated. Chain-of-custody AAS updated. SHACL validates integrity: package seal intact, temperature log shows no excursion, quantity matches shipment record.	21 CFR Part 312; GCP; ICH E6 R2	Layer 2 HL7/FHIR bridge → NiFi → RDF triple; Layer 3 SHACL integrity check; LKIF prohibition: IMP with broken chain-of-custody must not be dispensed
Subject Dispensing	Pharmacist records dispensing in EDC system. SWRL rule checks: subject eligible (protocol inclusion/exclusion), IMP not expired, dose calculation within protocol range, concomitant medication check.	GCP ICH E6 R2; Protocol compliance; 21 CFR 312.62	Layer 3 SWRL temporal rules; RDSG Step 2 temporal validation; LKIF obligation: pharmacist must confirm all dispensing checks before release
Biological Sample Management	Samples collected per protocol schedule. AAS SampleRecord created with subject ID (pseudonymized), collection timestamp, collection site, sample type, and storage conditions. CDISC	CDISC SEND, 21 CFR Part 11, GDPR pseudonymization	Layer 2 sample barcode scan → AAS SampleRecord; Layer 4 Clinical Trials Named Graph; LKIF obligation: subject pseudonymization before data enters knowledge graph

Stage	Action	Clinical Governance	Wymore / RDSG Control
	SEND ontology alignment.		
CRO Bioanalytical Data Return	CRO submits bioanalytical results via secure API (Kong Gateway). Kong validates mTLS certificate, checks data schema against CDISC standard. NiFi routes to HL7/FHIR bridge. SHACL validates result completeness.	ICH M10 bioanalytical method validation; CDISC SDTM	Layer 2 Kong API Gateway + HL7/FHIR bridge; Layer 3 SHACL completeness check; MTBH_WHAT monitors result concept resolution fidelity
Clinical Batch ↔ Lab Result Linkage	Knowledge Graph SPARQL query links ClinicalBatch → hasAnalyticalResult → BioanalyticalResult → forSubject → PseudonymizedSubject. AI agent can traverse entire genealogy for safety signal analysis.	ICH E2A pharmacovigilance; safety signal detection	Layer 4 SPARQL federated query across Clinical Trials + R&D Named Graphs; LangGraph Batch Investigation Agent
Clinical Knowledge Synthesis	Formulation Knowledge Synthesis Agent queries: 'Which batches of this formulation showed dissolution drift? What excipient lots were shared?' Cross-BL R&D intelligence.	ICH Q1A, Q10 continuous process improvement	Layer 6 LangGraph agent; Layer 5 SPARQL + SHACL; MTBH_WHY monitors intent-governance fidelity of synthesis queries

6.3 Flow C: Tech Transfer — R&D to Manufacturing

SEMANTIC DIGITAL THREAD: DISCOVERY → SCALE-UP → MANUFACTURING BOP → SAP BOM

This flow demonstrates how the holonic architecture enables seamless, semantically governed technology transfer from R&D to existing and future manufacturing sites, without requiring source system consolidation or manual translation of process parameters.

1. Tech Transfer Initiation: R&D scientist uses Mendix Tech Transfer Dashboard to initiate handover. SREG generates a tech transfer ORSD capturing the formal requirements: scale factor, target site capabilities, manufacturing equipment specifications, and regulatory filing requirements.
2. R&D Data Extraction: Because all R&D data was natively structured as RDF against the top-down ontology, the Knowledge Graph can immediately generate a tech transfer package. SPARQL query retrieves: FormulationDevelopmentBatch → hasParameters → ProcessParameters; → requires → EquipmentCapabilities; → produces → SpecificationSet.

3. SWRL Parameter Mapping: SWRL rules automatically map 'R&D Lab Parameters' to 'Manufacturing Line Constraints.' For example: LabScale_ImpellerSpeed_280RPM → maps_to → ManufacturingScale_AgitatorSpeed using the validated scale-up equation derived from Wymore performance specifications at both scales.
4. AAS-Based Site Matching: The AAS registry queries all manufacturing sites for EquipmentCapabilityMatch: which sites have bioreactors with the required volumetric capacity, agitation range, and GxP qualification status? AAS semantic normalization means a Sartorius bioreactor at Site A is directly comparable to a Applikon bioreactor at Site B.
5. SHACL Manufacturing Compliance Check: Layer 3 PEP validates the proposed manufacturing parameters against manufacturing site SHACL constraints: validated cleaning procedures, environmental monitoring classifications, equipment qualification status, personnel training records.
6. BOP Generation & Teamcenter Integration: Mendix pushes the harmonized data package directly into Siemens Teamcenter PLM as a unified Bill of Process (BOP). AAS submodels from Layer 2 provide the structured equipment specifications. KGCL change record created documenting the tech transfer.
7. SAP ERP BOM Update: Mendix updates SAP S/4HANA ERP with required raw material BOMs, production order structures, and cost center assignments. FCIDs (FABRIC Canonical Identifiers) maintain cross-system traceability between R&D batch records and manufacturing production orders.
8. Future Site Provisioning: For new manufacturing sites not yet operational, the AAS registry creates 'future state' shells populated with design specifications. When the site is commissioned and equipment is connected via IOX, the existing semantic model immediately populates with actual telemetry — no re-integration required.

6.4 Flow D: Continuous Improvement — SREG Feedback Loop

OPERATIONS CI REQUIREMENTS → SHACL UPDATES → PEP ENFORCEMENT → COMPLIANCE EVIDENCE

This flow demonstrates how the Wymore system model's self-evolution capability — the Observe-Reason-Adapt loop — is implemented through the SREG continuous improvement feedback mechanism:

- Observe: Manufacturing analytics surface a systematic yield variance of -3.2% across three sites on a specific formulation. Grafana OEE dashboard flags the deviation. Layer 4 SPARQL query identifies the correlated parameter: mixing time during buffer preparation.
- Reason: LangGraph Batch Investigation Agent traverses batch genealogy, LIMS results, and PI Historian data. Identifies that yield deviation correlates with buffer pH drift during mixing when ambient temperature > 22°C. Generates hypothesis: buffer preparation SHACL constraint needs tighter temperature control.
- SREG Intake: Operations CI team enters improvement requirement in AI4RE SREG: 'Buffer preparation ambient temperature must remain ≤ 20°C during pH adjustment step to prevent pH drift and yield variance.' AI4RE classifies and generates draft SHACL shape.
- Governance Review: Ontology Governance Committee reviews. Quality Manager confirms regulatory basis (ICH Q8 design space). LKIF obligation created. SHACL shape approved and deployed.
- PEP Enforcement: New SHACL constraint immediately active at Layer 3. Any subsequent batch failing the ambient temperature check is blocked from proceeding to pH adjustment step. HITL alert sent to production supervisor.

- Compliance Evidence: Full traceability from yield variance observation → CI requirement in SREG → SHACL shape deployment → PEP enforcement → corrective action audit trail. Regulatory Intelligence Hub surfaces this chain as ready-made FDA CPV evidence.

7. Modular Control Architecture & Scalability

The holonic architecture's modularity is its most strategically significant property. Because each holon is simultaneously self-governing (whole) and governed by its holarchy (part), the architecture can be extended to new domains, new manufacturing sites, and new clinical programs without re-engineering existing components.

7.1 Modular Control Design Principles

- Principle 1 — Holonic Autonomy: Each domain holon (R&D, Clinical, Manufacturing, Supply Chain) maintains its own SHACL constraints, SWRL rules, and PEP rule sets. A change in the Clinical holon's chain-of-custody constraints does not require re-validation of the R&D holon's analytical method constraints.
- Principle 2 — Protocol-Based Composition: Holons communicate through the RDSG governance protocol, not through shared schemas. A new manufacturing site is onboarded by creating its AAS shells and registering them with the AAS Registry — no changes to existing site integrations required.
- Principle 3 — Wymore State Isolation: Each holon's state space W is isolated. A state transition Δ in one holon does not directly modify W in another holon — all cross-holon propagation passes through the Layer 3 RDSG gateway.
- Principle 4 — SREG-Driven Governance Extension: New requirements from Marketing, Ops, or Compliance are absorbed by the SREG and formalized into new SHACL/LKIF artifacts without disrupting existing governance rules. The SHACL constraint library grows additively.
- Principle 5 — AAS Semantic Normalization: New equipment, sites, and materials are onboarded through AAS shell creation — not through custom integration code. Semantic equivalence across heterogeneous assets is governed by ECLASS 12.0 classification and IDTA submodel templates.

7.2 Scalability Patterns for R&D Expansion

Expansion Scenario	Onboarding Steps	Time to Governance	Existing System Impact
New R&D Lab Equipment Type	1. Create AAS NameplateV2 + TechnicalData shells. 2. Map OPC-UA signals to AAS submodel properties. 3. Register AAS in AAS Registry. 4. Add SHACL calibration constraint via VocBench. 5. Test PEP enforcement.	2–4 weeks	None — holonic autonomy isolates existing equipment
New R&D Site / Building	1. Deploy IOX local instance. 2. Configure OPC-UA adapters for site equipment. 3. Create ManufacturingSite AAS shell. 4. Register in AAS Registry. 5. Sync to FABRIC Knowledge Graph.	4–8 weeks (hardware-dependent)	None — AAS Registry is additive

Expansion Scenario	Onboarding Steps	Time to Governance	Existing System Impact
New Clinical Program	1. Create ClinicalTrial AAS with protocol metadata. 2. Configure HL7/FHIR bridge for CRO/EDC feeds. 3. Create InvestigationalMedicinalProduct AAS shells. 4. Deploy program-specific SHACL protocol compliance shapes. 5. Activate LangGraph Clinical Evidence Agent.	6–10 weeks (CRO integration dependent)	Clinical Trials Named Graph receives new program subgraph; no modification to existing programs
New Manufacturing Site (existing technology)	1. AAS shells created from equipment specifications (before site commissioning). 2. IOX deployed and connected to PLCs. 3. AAS shells populate with telemetry automatically. 4. Site-specific SHACL constraints (environmental monitoring classes) added. 5. Tech transfer flow available immediately.	8–16 weeks (includes IOX deployment)	AAS Registry updated; tech transfer SWRL rules automatically available for new site
New Manufacturing Technology (future state)	1. Technology-specific AAS submodel template created in DEV environment. 2. SWRL mapping rules developed for new process parameters. 3. SHACL constraints authored for new technology characteristics. 4. Validated in STG environment. 5. Promoted to PRD.	12–20 weeks	New domain holon created; existing holons unaffected
New Regulatory Jurisdiction	1. SREG Compliance intake: new regulatory requirements entered in AI4RE. 2. LKIF deontic norms formalized. 3. New SHACL shapes authored and reviewed. 4. PEP compliance domain updated with jurisdiction-specific rule set. 5. Regulatory Intelligence Hub updated.	4–8 weeks (regulatory review dependent)	New PEP compliance rule set added; existing rules preserved

8. Four-Domain Governance Framework

The FABRIC governance framework spans all seven architecture layers and provides the institutional structures, tooling, and enforcement mechanisms required for a compliant, trustworthy enterprise intelligence platform. The four domains are designed to interlock: semantic governance defines the concepts that data governance catalogs, analytics governance builds on the data that data governance validates, and AI governance monitors the models that analytics governance promotes.

8.1 Domain 1 — Data Governance

Tools: OpenMetadata · Apache Atlas · Great Expectations · OPA Policy Engine · SAP MDM

Capability	Implementation	R&D Application	Success Metric
Unified Data Catalog	OpenMetadata with automated discovery and schema profiling	All R&D datasets (ELN, LIMS, stability, clinical) catalogued with business glossary linkage	95% catalog coverage by Phase 2
End-to-End Lineage	Apache Atlas automated lineage: NiFi → dbt → Iceberg → Superset	Full provenance: SAP material master → NiFi extraction → dbt transformation → Knowledge Graph triple → analyst dashboard cell	100% production reports with end-to-end lineage by Phase 3
Data Quality Scoring	Great Expectations suites per domain; DQS calculated per dataset	R&D domain DQS threshold: >92%. Blocks promotion to Knowledge Graph on failure. Visible in OpenMetadata dashboard.	DQS >92% for R&D domain by Phase 1 completion
Master Data Management	SAP MDM for product, supplier, facility, and organizational master data	Single authoritative record for API/excipient/supplier master across all Baxter business lines	Cross-BL MDM coverage: 80% by Phase 2
PII & Sensitive Data Classification	OPA Policy Engine; attribute-based access control from ontological role definitions	Clinical trial subject data pseudonymized at Layer 2 before Knowledge Graph entry. GDPR compliance.	Zero PII exposure events in production
Data Stewardship	Designated stewards per domain with accountability metrics in OpenMetadata	R&D data steward owns formulation data domain; Clinical data steward owns trial data domain	Steward response time < 5 business days for escalations

8.2 Domain 2 — Semantic Governance

Tools: VocBench 3 · Protégé · OntoDB · SHACL Validator · Git (ontology repository)

Capability	Implementation	R&D Application	Success Metric
Centralized Ontology Registry	Git-backed versioning in VocBench 3. Every class, property, axiom tracked with KGCL change records.	R&D ontology terms (FormulationDevelopment Batch, StabilityStudy, AnalyticalMethod) version-controlled from ORSD to deprecation	Ontology concept coverage: 85% of critical business concepts by Phase 3
Controlled Vocabulary Management	SKOS-based domain vocabularies with lifecycle: proposed→review→approved→stable→deprecated	Clinical trial vocabulary aligned to CDISC SEND/SDTM; manufacturing vocabulary aligned to ISA-88/ISA-95	100% capability applications consuming governed ontology terms
SHACL Constraint Library	200+ shapes enforcing cross-domain data integrity invariants. Pre-deployment testing mandatory.	200+ shapes across R&D, Clinical, Manufacturing, Supply Chain, Regulatory Named Graphs	SHACL compliance rate >99.5% by Phase 3
Ontology Change Impact Analysis	Automated SPARQL query identifying all downstream artifacts affected by proposed changes	Before adding new OWL class, automated analysis shows which SHACL shapes, SWRL rules, and dashboards reference parent class	Impact analysis completed before every ontology deployment
External Ontology Alignment	OWL owl:equivalentClass and owl:sameAs mappings to IDTA/ECLASS, SNOMED CT, NCI Thesaurus, MeSH, FIBO, CDISC	Clinical trial concepts aligned to CDISC; manufacturing aligned to ECLASS 12.0 and VDI/VDE 3682	Key external alignments: ECLASS, CDISC, NCI Thesaurus (by Phase 2)
Ontology Governance Committee	Cross-functional bi-weekly review body with Architecture Board tie-breaking authority	R&D, Clinical, Manufacturing, Quality, Regulatory, Legal represented. AI4RE SREG outputs reviewed at each meeting.	Decision cycle ≤ 2 bi-weekly meetings per term dispute

8.3 Domain 3 — Analytics Governance

Tools: MLflow Model Registry · dbt Semantic Layer · Apache Superset RBAC · Great Expectations · Audit Logs

Analytics governance eliminates metric drift — the failure mode where 'batch yield' is calculated differently in different dashboards. The dbt Semantic Layer provides a single, version-controlled metric contract that all dashboards must reference. No ad-hoc SQL metric definitions permitted in certified dashboards.

Governance Mechanism	Implementation	R&D / Clinical Application
ML Experiment Registry	MLflow: every run tracked with hyperparameters, code version, data version, evaluation metrics	Stability outcome forecasting models, batch yield prediction, dissolution risk models — all tracked from first experiment to production
Model Promotion Workflow	experiment → candidate → staging (statistical validation) → production (MTBH-monitored, auto-rollback)	No R&D predictive model reaches production without MTBH baseline established and Circuit Breaker configured
Metric Source of Truth	dbt Semantic Layer: canonical metrics (batch yield, specification compliance rate, cycle time) version-controlled	Cross-BL formulation yield defined once in dbt; all dashboards reference same metric — drift structurally impossible for certified views
Dashboard Certification	'Baxter Certified' badge on dashboards passing data quality, business logic, and peer review gates	R&D Intelligence Platform dashboards certified before scientist access; Self-Service outputs watermarked 'Unvalidated'
Role-Based Access via Ontology	Ontological role assignments from Knowledge Graph drive data access — no manual permission matrix	Clinical Trial data accessible only to personnel with ClinicalTrialRole and current GCP training record in ontology
Analytics Audit Log	All dashboard views, report exports, model inferences logged for regulatory readiness	21 CFR Part 11 compliant audit log; FDA inspection-ready export from Superset and MLflow

8.4 Domain 4 — AI Governance (MTBH Framework)

Tools: MTBH Circuit Breaker · OPA Policy Engine · Model Cards · EU AI Act Readiness Scanner · Immutable Audit Log

The MTBH Framework operationalizes AI trustworthiness as an engineering discipline. The Mean Time Between Hallucinations metric provides a formal, quantitative trust measure for all generative AI and ML model deployments. The Circuit Breaker enforces automatic model suspension before unreliable outputs reach Baxter users.

Governance Capability	Implementation Detail	R&D / Clinical Governance
MTBH Per Model	Calculated per deployed model per deployment window. Updated continuously. Surfaced in model registry and capability app health dashboards.	Patient-safety-adjacent use cases (clinical batch AI) require higher MTBH than operational optimization (yield prediction)
Circuit Breaker Architecture	Automated model suspension when MTBH falls below use-case-specific threshold. Zero-downtime failover to last stable model.	If Batch Investigation AI confidence falls below threshold during safety-critical deviation analysis, immediate suspension + HITL mandatory

Governance Capability	Implementation Detail	R&D / Clinical Governance
Composition Theorem Enforcement	Multiplicative fidelity degradation tracked across multi-model pipelines. Prevents compounding individual model uncertainty.	LangGraph multi-step agents: each hop in the reasoning chain reduces composite fidelity. Threshold adjusted for pipeline length.
Sanitization Framework	Input and output validation for all LLM inferences. Tripped Circuit Breaker outputs excluded from training data (anti-collapse mechanism).	Failed clinical batch analysis outputs never contaminate training data. KGCL quarantine record created for each tripped output.
Model Cards	Bias assessments, intended use declarations, known limitations, out-of-scope warnings. Published to MLflow model registry.	Regulatory Evidence Agent model card: intended for CMC dossier assistance; not approved for standalone regulatory decision-making
HITL Gates	Mandatory human review for all AI outputs with patient safety implications. No autonomous action on regulated decisions.	Any AI recommendation affecting IMP dispensing, clinical batch release, or manufacturing process change requires documented human authorization
EU AI Act Compliance	Article 9 risk classification per AI system, technical documentation, post-market monitoring plan.	Clinical batch AI systems classified as high-risk AI under EU AI Act. Technical documentation maintained in Regulatory Graph.

9. Monitoring, Reporting & Observability

The holarchic observability framework — the Tom Sawyer Perspectives-enhanced OPS Workbench — surfaces governance state in real time across all four holonic levels. It is not a reporting tool; it is the holarchy's self-awareness system.

9.1 Technical Platform Observability

Metric Category	Tool	Key Metrics	Alert Threshold
Knowledge Graph Health	Stardog + custom SPARQL probes	Triple count growth rate, SPARQL query response time, SHACL validation throughput, reasoning latency, named graph replication lag	SPARQL P95 > 2s; SHACL failure rate > 0.5%
Integration Pipeline Health	Apache NiFi + Kafka + Grafana	NiFi flow throughput (records/min), Kafka consumer lag (per topic), Airbyte sync success rate, integration error rate, DLQ queue depth	Kafka lag > 10K messages; NiFi error rate > 1%
AAS Registry Health	Siemens IOX + custom probes	AAS shell creation rate, submodel sync latency, OPC-UA signal ingestion success rate, AAS-to-triple conversion throughput	OPC-UA ingestion failure rate > 2%; AAS sync lag > 30s
AI Model Performance	MLflow + Ray Serve + MTBH service	MTBH score per model (all 5 dimensions), Circuit Breaker trip count (per period), inference latency P95, model accuracy (rolling 20-decision window)	MTBH below threshold → immediate suspension; trip count > 3/day
Human Performance	Custom Node.js service + PostgreSQL	Fatigue score distribution, trust mis-calibration (by confidence bucket), decision accuracy trend, production pressure score (team-level)	Fatigue score > 0.7; team pressure score > 0.7; accuracy drop > 15%
Data Quality	Great Expectations + OpenMetadata	DQS per dataset per domain, expectation failure rate, data promotion block rate, catalog coverage percentage	DQS < 80% blocks promotion; DQS < 90% for R&D triggers steward alert

9.2 Business Intelligence & Governance Dashboards

Dashboard	Tool	Audience	Key Content
R&D Intelligence Platform	Apache Superset + LangGraph NLQ	R&D Scientists, Formulation Managers	Cross-BL formulation query results, stability trend charts, failed experiment knowledge, prior art synthesis, MTBH scores for R&D AI agents
Batch Intelligence & QC Release	Superset + Mendix HITL	QC Scientists, QA Managers	Real-time batch status, LIMS-MES semantic linkage, QC release decision

Dashboard	Tool	Audience	Key Content
			support, batch genealogy visualization, HITL pending queue
Regulatory Intelligence Hub	Superset + LangGraph Regulatory Agent	Regulatory Affairs, CMC	CMC traceability from development to submission, specification version history with full audit trail, regulatory change impact analysis, eCTD section completion
Manufacturing Process Analytics	Grafana + Superset	Manufacturing Engineers, Plant Managers	OEE, cycle time, yield, CPV dashboards; equipment performance trending; process deviation detection with Knowledge Graph cross-reference
MTBH & AI Governance	Custom Grafana dashboards	AI Governance Lead, Data Science	MTBH scores per model (all 5 dimensions), Circuit Breaker trip history, model promotion status, EU AI Act compliance score
Human Performance Monitor	Custom Mendix dashboard	Knowledge Engineering Managers	Team fatigue heat map, trust calibration distribution, production pressure score, competency assessment results, mandatory break compliance
Governance & Compliance Command Center	Superset + OpenMetadata	Quality Directors, Compliance, IT	SHACL compliance rate trend, ontology coverage, data quality scores, PEP violation log, HITL response time, 21 CFR Part 11 audit readiness
Executive Program Health	Superset + custom scorecard	CTO, CDO, VP R&D, VP Manufacturing	Platform TCO vs. plan, FABRIC adoption rate (weekly active users), capability app uptake, phase gate status, value realization vs. business case

10. Implementation Roadmap

The implementation is organized across three phases over 36 months. Each phase builds on the preceding phase's infrastructure and organizational capability. The phased approach delivers measurable value within 12 weeks while building the institutional infrastructure required for enterprise-scale deployment.

10.1 Phase 1 — Foundation (Months 1–6): R&D Intelligence & Core Infrastructure

Month (s)	Workstream	Key Deliverables	Owner
1	AI4RE SREG Setup	AI4RE tool deployed; SREG intake workflows for Marketing, Ops, and Compliance; initial ORSD templates created	TVE Semantic Architect
1	Infrastructure	Cloud environment, network security zones, CI/CD pipeline, GitOps repository, Kafka cluster, AAS Registry deployment	Platform Architect
1–2	Layer 1–2 (R&D)	SAP LIMS/ELN integration via NiFi and Airbyte; Kafka topics for R&D domains; IOX pilot deployment at R&D facility; first AAS shells for lab equipment	Integration Lead + Siemens
1–2	Wymore Model M	Formal Wymore system model documented for all seven layers; R-specifications (performance specs) defined for R&D domain; SHACL shapes derived from R-specs	Enterprise Architect
2–3	Layer 4 (Data Fabric)	Iceberg table design for R&D domain; dbt models for ELN, LIMS, SAP R&D objects; Great Expectations suites (DQS target >90%)	Data Engineer
3–4	Layer 5 (Knowledge Graph)	Seed ontology from IDTA/ECLASS; OWL 2 class hierarchy for R&D domain; 50+ SHACL shapes; SPARQL endpoint live; SWRL rules for stability/specification temporal reasoning	Knowledge Engineer
3–4	Layer 6 (Mendix)	HITL Intervention Pane; SHACL Validation Viewer; Ontology Context Explorer; AI4RE Requirements Dashboard; Human Performance Monitor baseline	Mendix Developer
4–5	Layer 7 (R&D Intelligence)	R&D Intelligence Platform MVP: cross-BL formulation query, ELN harmonization view, failed experiment search; Batch Intelligence MVP: LIMS-MES linkage, batch genealogy, QC release support	Product Owner: R&D
4–5	MTBH Baseline	MLflow deployed; JupyterHub operational; MTBH service deployed; baselines set for all R&D AI models; Circuit Breaker configured for R&D domain	Analytics Lead
5–6	Human-AI Teaming	Real-time performance monitoring live; trust calibration active; cognitive workload	Human Factors Lead

Month (s)	Workstream	Key Deliverables	Owner
		management deployed; fatigue detection and break enforcement operational	
6	Phase Gate 1	20+ weekly active users on R&D Intelligence; DQS >90% for R&D domain; Governance baseline approved; Phase 2 funding authorized by Steering Committee	Program Director

10.2 Phase 2 — Scale (Months 7–18): Manufacturing, Clinical & Supply Chain Expansion

Quarter	Workstream	Key Deliverables	Value Delivered
Q3 (M7–9)	Clinical Trials Integration	HL7/FHIR bridge for CRO feeds; ClinicalTrialMaterial AAS shells; Clinical Trials Named Graph; chain-of-custody SHACL constraints; IMP dispensing SWRL rules	Full clinical material governance from IMP release to bioanalytical data return
Q3 (M7–9)	Manufacturing Integration	Opcenter MES, PI Historian, SCADA via Siemens IOX; manufacturing NiFi flows; manufacturing batch genealogy graph; CPV dashboards	Manufacturing data domain flowing into FABRIC
Q3–Q4 (M8–12)	AAS Expansion	AAS shells for all manufacturing assets across all sites; full-facility digital twin coverage; site-agnostic semantic normalization operational	Cross-site analytics without ERP/MES consolidation
Q4 (M10–12)	Tech Transfer Flow	Tech Transfer Dashboard in Mendix; SWRL parameter mapping rules; BOP generation to Teamcenter; SAP BOM auto-update; future-site AAS provisioning	Semantic digital thread from R&D to manufacturing
Q4–Q5 (M10–15)	Capability Apps 3 & 4	Manufacturing Process Analytics MVP; Supply Chain Risk Observatory MVP; Clinical Trial Material Tracker MVP	Cross-domain intelligence across all major BLs
Q5 (M13–15)	Portfolio Rationalization Wave 1–2	Standalone BI tools consolidated to Superset; shadow data warehouses migrated to Iceberg; duplicate ontology tooling consolidated to VocBench; decommissions Wave 1–2 complete	\$450K–\$1.05M annual savings
Q5–Q6 (M14–18)	Capability Apps 5 & 6	Quality & Compliance Command Center MVP; Regulatory Intelligence Hub Phase 1; Process Automation Workflows MVP	GxP compliance dashboard and regulatory evidence traceability
M18	Phase Gate 2	All 6 capability apps at >60% weekly active adoption; supply chain digital twin covering 80%+ Tier 1 suppliers; MTBH active for all production AI models; Wave 1–2 decommissions confirmed	Phase 3 planning authorized

10.3 Phase 3 — Optimize (Months 19–36): AI Augmentation & Full Governance Maturity

Timeline	Theme	Key Activities	Strategic Outcome
M19–24	AI Augmentation	LangGraph agent deployment for batch investigation, formulation synthesis, regulatory template generation; MTBH threshold tuning per use case risk class; FLAML predictive models in production	AI-augmented R&D and manufacturing workflows with MTBH-guaranteed fidelity
M21–27	Regulatory Intelligence	SPL/eCTD ontology; CMC traceability graph; Veeva RIM integration; FDA/EMA regulatory change monitoring; Regulatory Evidence Agent; EU AI Act Article 9 compliance documentation	Automated CMC traceability; eCTD population assistance; regulatory AI compliance posture
M24–30	Knowledge Federation	AAS-based federated integration with CMO partners and Tier 1 API suppliers; SPARQL federation endpoint; partner data quality SLA agreements; Wymore model extended to partner holons	Ecosystem-level supply chain intelligence without direct system access
M27–33	Governance Maturity	Self-service ontology extension toolkit; predictive data quality monitoring; EU AI Act compliance documentation; FDA AI/ML predetermined change control protocols; full LKIF deontic norm library	Full governance maturity (Level 3 all four domains); regulatory AI compliance posture
M30–36	Optimization	Legacy decommission Wave 3 (SAP BW reduction, reporting databases); infrastructure cost optimization; advanced CPV analytics; cross-BL benchmarking suite; Wymore model performance specification refinement	Full TCO target; \$720K–\$1.71M annualized rationalization savings

11. Risk Register

Risk	Category	L	I	Mitigation Strategy
AI4RE SREG adoption resistance from Marketing/Ops/Compliance requirement owners	Organizational	M	M	Dedicated change management program. AI4RE trained with existing requirement templates. Demonstrable time-to-governance reduction shown in first 6 weeks of use. Executive sponsor mandate for SREG as single intake channel.
Wymore Model M formalization delays architecture delivery	Technical	M	M	Wymore formalization is a documentation activity that runs parallel to implementation. Formal model is refined iteratively — first version complete by Month 2; refined through Phase 1 as implementation reveals gaps.
Source system access restrictions delay integration timelines	Technical / IT	H	H	Formal data access governance agreements in Month 1. Pre-negotiate read-only service accounts for all 10 priority source systems before kickoff. CTO escalation if access blocked >10 business days.
Siemens AAS adoption complexity greater than modeled	Technical	M	H	Begin with AAS shells for 10–15 highest-value asset types only. Engage Siemens Engineering Services from Month 1. Pre-built pharma AAS templates reduce effort by 60%.
Human-AI Teaming framework adoption resistance from knowledge engineers	Organizational	M	M	Frame as engineer support and protection tool — not surveillance. Engineers involved in design of trust calibration UI. Performance monitoring data shown only to engineers themselves and their managers.
MTBH thresholds mis-calibrated for clinical safety-critical use cases	AI / Quality	L	H	MTBH thresholds calibrated conservatively in Phase 1 for all patient-safety-adjacent use cases. Clinical pharmacist and medical officer review all MTBH configurations for clinical trial AI applications. Independent audit before any clinical AI goes to production.
Ontology Governance Committee reaches stalemate on terminology	Organizational	M	M	Pre-define ontology decision protocol (lazy consensus with 5-day comment window). Architecture Board has explicit tie-breaking authority. Time-box term debates to 2 bi-weekly meeting cycles before escalation.
GxP validation requirements for FABRIC impose compliance burden	Compliance	M	H	FABRIC read-only integration avoids GxP validation burden on source systems. FABRIC itself follows Computer System Assurance (CSA) approach per FDA 2022 guidance. Engage Baxter Compliance in Month 1 to agree CSA scope.
Automation complacency degrades human oversight of AI outputs	Human Factors	H	H	Human-AI Teaming Framework implemented concurrently with AI deployment. MTBH Circuit Breaker prevents low-confidence outputs reaching users. Competency maintenance program (bi-weekly manual mapping exercises) preserves override capability.

Risk	Category	L	I	Mitigation Strategy
Key talent shortage — knowledge engineers and semantic architects	Resourcing	M	H	TVE Group provides knowledge engineering for Phase 1–2. Parallel training of 2 Baxter internal staff in OWL/SHACL from Month 1. Stardog/GraphDB professional services as backup resourcing contingency.

12. Success Metrics & KPIs

KPI	Baseline	Phase 1 Target	Phase 2 Target	Phase 3 Target
R&D cross-BL formulation query time	Days (manual)	<30 minutes	<5 minutes	Real-time (seconds)
Clinical material chain-of-custody completeness	~60% traceable	100% new programs	100% all programs	100% + digital audit export
Tech transfer cycle time (R&D → Mfg BOP)	8–12 weeks manual	4–6 weeks (semantic-assisted)	2–3 weeks	<2 weeks (automated SWRL mapping)
SHACL compliance rate	N/A	Baseline established	>97%	>99.5%
Data catalog coverage	~10%	40%	75%	95%
Weekly active FABRIC users	0	20–30 (R&D pilot)	150–200	400+
MTBH score (primary AI models)	N/A	Baseline defined per model	Within threshold 100% of window	Continuous monitoring; all thresholds met
Human-AI Trust Mis-calibration	N/A	Baseline established	<10% delta from actual accuracy	<5% delta from actual accuracy
Validation accuracy (knowledge engineers)	N/A	>90% all sessions	>92%	>95% with framework assistance
EU AI Act readiness score	N/A	Assessment complete	Remediation plan active	>85% compliant
Platform TCO vs. Palantir equivalent	Benchmark only	Infrastructure baseline	\$6–10M savings demonstrated	\$6–13M/yr sustained savings
AI4RE SREG requirements throughput	0 (not deployed)	30+ formalized requirements	100+ requirements; all three source domains active	Self-service SREG for business users
Legacy systems decommissioned	0	2–3 (tactical tools)	4–6	8–10+
Wymore R-specification compliance	N/A	R-specs defined for R&D domain	R-specs defined all domains; 95% transitions compliant	100% transitions formally validated

13. Document Information & Governance

Document Title	Baxter Healthcare Holonic Semantic Architecture — Unified Detailed Design Document v2.0
Version	2.0 — DRAFT for Stakeholder Review
Date	May 2026
Prepared by	The Value Enablement Group, LLC
Primary Author	Jeffrey Walkk, Senior Enterprise Architect & Knowledge Engineering Leader
Architecture Framework	Requirements-Driven Semantic Gateway Architecture (RDSG v2.0) + Wymore System Model M + AI4RE SREG
Classification	CONFIDENTIAL — Baxter Healthcare and TVE Group internal use only
Source Documents Integrated	Baxter DDD (Holonic Semantic Architecture); Baxter FABRIC Solution Architecture v1.0; Holonic Semantic Governance Architecture v2.0; Human-AI Teaming Framework (February 2026)
Next Review Date	Upon stakeholder review feedback incorporation
Intended Audience	Baxter Executive Steering Committee, CTO Office, VP R&D, VP Manufacturing, VP Clinical, IT Leadership, Architecture Board

Source Framework Integration Summary

Source Framework	Primary Contribution to This Document	Sections Referenced
Baxter DDD (Holonic Semantic Architecture)	7-layer holonic model; tool recommendations; HITL governance flows; R&D-to-manufacturing operating flows	Sections 4, 5, 6
Baxter FABRIC Solution Architecture v1.0	7-layer FABRIC architecture; four-domain governance; implementation roadmap; PMO; risk register; KPIs; portfolio rationalization	Sections 4, 5, 8, 10, 11, 12
Holonic Semantic Governance Architecture v2.0	Wymore Model M integration; holonic theory (Koestler + Cagle); RDSG v2.0 six-stage traversal; MTBH Circuit Breaker; ORSD/KGCL; SUDA; RAG; standards foundation	Sections 2, 3, 5, 6, 7, 8
Human-AI Teaming Framework (Feb 2026)	Automation paradox analysis; human performance monitoring; trust calibration; cognitive workload management; competency maintenance; production pressure detection	Sections 5.2, 8, 9, 11, 12
Wymore System Model M (Added)	Formal mathematical system definition; state space (W), transition function (Δ), performance specifications (R); formal layer boundary contracts	Sections 2, 5 (all layers), 6, 7, 12

Source Framework	Primary Contribution to This Document	Sections Referenced
AI4RE Semantic Requirements Engineering Gateway (Added)	Requirements intake from Marketing, Ops, Compliance; AI4RE processing pipeline; SREG-to-RDSG integration; continuous governance feedback loop	Sections 3, 5.3, 6.4, 8

— End of Document —

The Value Enablement Group, LLC | Jeffrey Walk | Version 2.0 | May 2026
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